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Air Quality Index Prediction and Attention Faithfulness Across Eight Indian Cities

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***Abstract*—Air quality forecasts let people and authorities act before pollution peaks, and attention-based models are often presented as showing which inputs drive those forecasts, yet this is rarely tested. In this study, we examine station-level Air Quality Index (AQI) on a daily basis from 2023 through 2026 at 107 CPCB stations in eight Indian cities. The station data are augmented with satellite, reanalysis, and forecast-weather data, and the performance of ridge regression, GRU and XGBoost are tested in the year-level lock test, and the attention mechanism is analyzed with respect to SHAP through input-replacement sensitivity analysis. All of the models are benchmarked against a baseline based on the previous day's AQI. In 2025, the 28-day GRU model reduces the RMSE from 35.388 to 32.516, while the weather-enabled XGBoost model reduces it from 34.668 to 30.270 in 2026, where most of the reduction is thanks to weather forecast. Attention helps, harms, and makes no difference in 11, 11, and 14 out of 36 pairs of tests, respectively, and attention always disagrees with SHAP more often than input-replacement. Attention maps should be checked before being used for explanation. An out-of-time evaluation begins in January 2027.**

***Index Terms*—air quality index, attention faithfulness, explainable AI, time-series forecasting, GRU, XGBoost, SHAP, cross-city transfer**

I. Introduction

A. Background and importance

The pollution of air is one of the biggest environmental threats to human health. According to the estimates of the World Health Organization, the air pollution resulted in 4.2 million premature deaths globally in 2019; 99% of the global population were exposed to the air which did not meet its air quality guidelines; also, 89% of the mentioned deaths happened in low- and middle-income countries [1]. 8.1 million deaths caused by air pollution were stated by the State of Global Air report in 2021 [2]. India carries a large share of this burden: the Global Burden of Disease study for Indian states attributed 1.67 million deaths in 2019, 17.8% of all deaths in the country, to air pollution, with an economic loss of US$36.8 billion, or 1.36% of gross domestic product [3].

B. Problem context and practical significance

The Air Quality in India is assessed based on the Air Quality Index (AQI) derived from CPCB monitoring stations [4]. Based on the data for this particular project, 20.8% of station-days between 2023 and 2025 were rated as being either Poor, Very poor or Severe, meaning that almost one out of every five station-days was unhealthy. A day-ahead forecast gives people an opportunity to respond to the predicted spike in pollution; however, this forecast can be considered helpful only if it is better than the most obvious baseline (i.e., tomorrow will be like today) and only if its inputs are trustworthy.

Two pain points follow. First, tomorrow's AQI is strongly related to today's, so a model can look accurate while adding little; in this study the best AQI-history forecaster's output correlated 0.989 with yesterday's value. Second, attention models are routinely explained by showing their attention weights, but it has not been established that those weights reflect what drives the output of an AQI regression model.

C. Research gap

The critical gap is that attention is adopted as an explanation without a fixed-model test. Indeed, the discussed air quality studies demonstrate improvements in predictive power and ablation study of components for attention [5], [6], [7], comparison of explanation approaches in classification datasets or single monitoring networks [8], [9], [10], but none within the scope of our literature survey [11] offers an evaluation of a fixed-model attention faithfulness test, group of realistic perturbations, and consistency between cities of station-level multimodal AQI regression. The reason is that accuracy does not indicate an incorrect explanation: one might believe that the model takes into account wind or satellite features when it actually neglects them. The issue of attention as explanation

was raised in text classification [12], [13], and its solution for correlated continuous environmental inputs is currently unknown. Another obstacle is related to evaluation methodology, since the literature survey differs in splits, targets and horizons, and in one case, the independence of the split from preprocessing remains undetermined [14].

D. Research questions

1) How much do a history-based model and a weather-assisted model improve next-day station AQI forecasts over the previous-day baseline in locked chronological tests?
2) Does attention provide any improvement over uniform-weighting in terms of accuracy, and does this vary depending on the input used?
3) Do attention rankings agree with measured input-replacement sensitivity as well as SHAP rankings do?
4) Which inputs account for the forecast skill beyond the baseline?

E. Research objectives

1) To build, by September 2026, an aligned daily dataset for 107 CPCB stations in eight cities with a verified end-of-day AQI target, satellite and weather inputs and documented quality flags.
2) To evaluate ridge, GRU and XGBoost forecasters in three locked tests (2024, 2025 and January–September 2026), each scored once against the baseline with 95% paired block-bootstrap intervals.
3) To compare 36 attention variants with uniform controls on two input sets under chronological, leave-one-city-out and season-block designs.
4) To audit attention and SHAP rankings against replacement sensitivity on 300 sampled rows per checkpoint.
5) To quantify, with input ablations and importance scores, how much forecast skill comes from yesterday's AQI, older history, calendar and location, and weather forecasts.
6) To publish the saved results in a dashboard that separates same-day estimates, backtests and the pending out-of-time evaluation.

F. Research contribution

The paper has four contributions. First, it offers three locked and independently validated forecast evaluations relative to a clear-cut previous-day baseline for eight Indian cities. Second, it offers pairings on 36 attention types in which the accuracy of attention is contingent on the nature of the input set and testing. Third, it performs audits of attention using SHAP with replacement sensitivity, and finds that agreement between the two methods is always less in each audit. Fourth, it reveals the small amount of forecast skill derived from AQI history other than that of yesterday, and the amount from weather forecasts.

G. Paper roadmap

The second section covers literature review and gap. The third section highlights methodology. Results and discussions are covered in the fourth section. Section five covers conclusions while Section six talks about limitations of the study. Section seven covers future scope.

II. Related Work

A. Classification of approaches

The ten selected papers [11] have been divided into three overlapping categories, while the attention problem and the techniques of post-hoc explanations constitute the context of this research. The attention mechanism was proposed as a soft alignment technique [15], and it has been proven that attention weights can be changed without affecting the predictions [12], provided that the definition of explanation is chosen accordingly [13]. Post-hoc explanation, such as SHAP, assigns contribution to every input to a prediction [16], however, different explanation methods may contradict each other on the same model [17].

Explanation evaluation and attribution studies tested explanations against prediction changes [8], [9] or made grouped attribution efficient [18]. *Attention-based prediction and spatial assessment* studies used attention for pollutant or AQI prediction [19], [10], [14], [6], [7]. *Transfer learning and spatial generalization* studies examined prediction at sites with little data [5], [20]. Scores in these papers use different targets, horizons, data and splits, so they cannot be ranked against one another, and AQI differs from PM2.5 and PM10 concentrations.

TABLE I
COMPARATIVE SUMMARY OF THE REVIEWED STUDIES

Study	Year	Method	Key contribution	Study limitation
Turbé et al. [8]	2023	Compared six attribution methods on CNN, BiLSTM and Transformer classifiers with corrupted blocks	Corruption-based audit that ranks attribution methods	Transfer to regression remains unvalidated.
Šić et al. [9]	2025	Tested 23 perturbation methods and 12 attribution methods on five UCR datasets	Showed that attribution rankings depend on the perturbation design	The metrics are limited to classification probabilities.
Ma et al. [5]	2024	Pretrained an adaptive bidirectional GRU with temporal and external attention on nine Beijing stations, then fine-tuned on the target site	Two-stage attention for sites with little data	Adaptation requires target labels.
Li et al. [19]	2023	Trained a physics-inspired graph network with attention on 1,604 stations in China	Combined transport physics with learned spatial relations (site-based PM10 R^2 about 0.85)	The physical meaning of the learned coefficients is limited.
Zhang et al. [10]	2026	Trained an explainable Transformer (X2-AQFormer) for NO_x and PM10 in Stockholm up to 72 hours ahead	Intrinsic explanations compared with GradientSHAP	The evaluation is confined to the Stockholm network.

Liu et al. [14]	2025	Combined a Transformer and a BiLSTM to predict daily AQI in three northern Chinese cities	Daily AQI forecasting with SHAP feature analysis	The preprocessing description does not establish that the split is independent of fitting.
Poelzl et al. [20]	2025	Compared predictors, generalization and adaptation (TrAdaBoostR2) for PM10 between Graz and Zagreb	Separated generalization from label-assisted adaptation	The experiment covers two cities and excludes exceptional events.
Hertel et al. [18]	2026	Introduced SHAPformer, a masked-attention Transformer, on electricity data	Sampling-free SHAP for grouped inputs	The cost grows exponentially with the number of feature groups.
Chen et al. [6]	2023	Proposed STA-ResConvLSTM for PM2.5 around Beijing	Spatial–temporal attention in residual blocks	The model is limited by abrupt changes and by horizons beyond 48 hours.
Liu et al. [7]	2023	Developed STAA-GCN for PM2.5 on Beijing, Tianjin and London datasets	Adaptive graph attention over station relations	Prediction in unmonitored areas is outside the evaluation.

B. Gap synthesis

The reviewed literature shows that attention is widely used for air-quality prediction, but the evidence for it comes in three kinds that are rarely combined: predictive gains and component ablations [5], [6], [7]; comparisons of explanation methods on classification data [8], [9] or one monitoring network [10]; and transfer experiments that mix zero-shot, station-holdout and label-assisted settings [5], [19], [14], [20], [7]. These create four gaps. First, there is the methodological gap of no paper in the review using attention against the input replacement sensitivity test on the fixed model for station-level multimodal AQI regression. The knowledge gap is that the discussion of whether attention can be used to explain the model has not been used with correlated inputs. The population gap is that the studies have focused on one or two cities only, hence there is lack of uniformity across the regimes. There is also the empirical gap that there is no solution to conflicting ideas about attention being faithful in this particular case. These are limitations of our literature review and not proof of non-existence of such a paper. (Table II).

TABLE II
MAPPING OF RESEARCH GAPS TO OBJECTIVES AND CONTRIBUTIONS

Gap type	Addressed by	Objective	Section
Methodological	Fixed-model replacement audit of attention versus SHAP	4	III-H, IV-E
Knowledge	36 paired attention-versus-uniform comparisons on environmental inputs	3	IV-D
Population	Eight cities, leave-one-city-out and season-block designs	1, 3	III-A, IV-D
Empirical	Baseline comparison and input ablations on locked tests	2, 5	IV-A to IV-C

III. METHODOLOGY

A. Dataset description

Hourly station AQI came from CPCB exports [4]. Satellite inputs were Sentinel-5P NO_2 and HCHO [21] and MODIS aerosol optical depth [22], and weather inputs were ERA5-Land daily reanalysis [23], all extracted through Google Earth Engine on a 0.03° grid. The 2026 weather model also used GFS forecasts issued 24 hours before each valid hour, taken from the Open-Meteo Previous Runs API [24]. CPCB is the national agency for the official index, and the satellite and reanalysis products cover every city, which made one consistent pipeline possible. The 107 stations are in Delhi (39), Hyderabad (14), Mumbai (14), Bengaluru (13), Chennai (8), Kolkata (7), Jaipur (6) and Lucknow (6), covering plains, coastal, upland, semi-arid and delta climates (Table III). A further 20,086 station-days from January to September 2026 at 91 stations form the weather test.

TABLE III
ALIGNED DATA BY YEAR

Year	Station-days	With AQI	Stations
2023	31,813	30,081	107
2024	33,860	31,722	106
2025	34,189	32,842	107
All	99,862	94,645	107

The target is skewed. Of 93,815 usable station-days, 12.4% were Good (AQI 0–50), 35.1% Satisfactory (51–100), 31.6% Moderate (101–200), 11.0% Poor (201–300), 7.4% Very poor (301–400) and 2.4% Severe (401–500). The mean was 138.4, the median 105 and the standard deviation 98.4.

B. Preprocessing

1) *Time alignment.* CPCB's hourly AQI is a trailing 24-hour average, so a daily mean is blurred across two days: its correlation with the previous day's satellite NO_2 (0.217) nearly matched that with the same day's (0.230). The target is *AQI_eod*, the hourly AQI at 05:00 IST on the following day, whose window covers almost exactly the UTC day described by the inputs. All dates are UTC.
2) *Cleaning.* Days on which AQI stayed constant below the 500 cap were flagged as stuck-sensor readings (906 rows) and excluded. Missing targets were never imputed. An independent audit reproduced all 31,813 aligned 2023 rows from the raw files with no mismatches.
3) *Missing inputs.* Satellite values are kept as missing under cloud. Tree models read missing values natively; ridge used median imputation with missing indicators; the GRU received a validity channel and zeros after normalization.
4) *Normalization.* Inputs were standardized with the training-window mean and standard deviation (z-score), which suits gradient-trained networks and keeps the scale of each input comparable. The GRU target was standardized the same way.
5) *Encoding.* Day of year was encoded cyclically as sine and cosine so that 31 December and 1 January are close. City and station identity are not inputs; location enters only as latitude and longitude.
6) *Balancing.* No resampling was used because AQI is continuous and resampling would distort the target. The

effect of the skew is instead exposed by reporting errors per AQI band.

C. Feature engineering and selection

Table IV lists the input groups. For next-day forecasts every input for target day D uses only observations from days up to D−1, and weather forecasts were issued before D began. Three feature families were compared for the tree and linear models: a short family of 9 inputs, a long family of 45 and a delayed-weather family that adds ERA5 values lagged 7 and 14 days (57 inputs). There was no importance-based pruning; the family and model were chosen by lowest RMSE on October 2024, a month held out from fitting (Table V).

TABLE IV
INPUT GROUPS

Group	Inputs	Count	Used by
Past AQI	Previous 28 daily AQI values	28	GRU, XGBoost, ridge
AQI summaries	Mean, spread and valid-day count over 3, 7, 14, 28 days; change from the day before	13	XGBoost, ridge
Calendar	Day of year (sine, cosine)	2	All models
Location	Station latitude, longitude	2	All models
Forecast weather	GFS temperature, humidity, wind speed, rain, pressure, issued 24 h ahead	5	Weather XGBoost
Satellite and ERA5	NO_2, HCHO, aerosol; temperature, dew point, wind, pressure, rain (same day)	n/a	Same-day ridge, attention study

TABLE V
MODEL SELECTION ON OCTOBER 2024 (2,642 ROWS)

Candidate	Inputs	RMSE
GRU-28days	28-day window	31.501
Ridge-short	9	33.483
XGBoost-delayed ERA5	57	33.624
RandomForest-delayed ERA5	57	34.036
RandomForest-long	45	34.350
Ridge-delayed ERA5	57	34.504
Persistence	yesterday only	34.546
Ridge-long	45	34.673
XGBoost-long	45	34.881
XGBoost-short	9	35.358

D. Algorithm selection and justification

The baseline repeats the previous day's AQI, which every candidate must beat. Ridge regression is a transparent linear reference. XGBoost [25] fits non-linear trees, handles missing values and was trained on the change from yesterday so that the baseline is its default answer. GRU [26] is a recurrent neural network which uses a 28-day lookback and is capable of utilizing time-dependent lagged features which can only be captured partially. A three-seed GRU ensemble reduces seed variation. In Table V, the GRU obtained the best performance in terms of RMSE on the selection month, outperforming all tree and linear models and was also frozen for the test year 2025. For 2026, another candidate model used archived weather forecasts combined with XGBoost. The attention-based model used GRU encoder with either temporal, feature or spatial attention as their weights were explainable.

E. System workflow

Fig. 1 shows the workflow from raw data to dashboard. The chronological split is the central safeguard: the model is fitted, selected and calibrated before the test period is scored once.

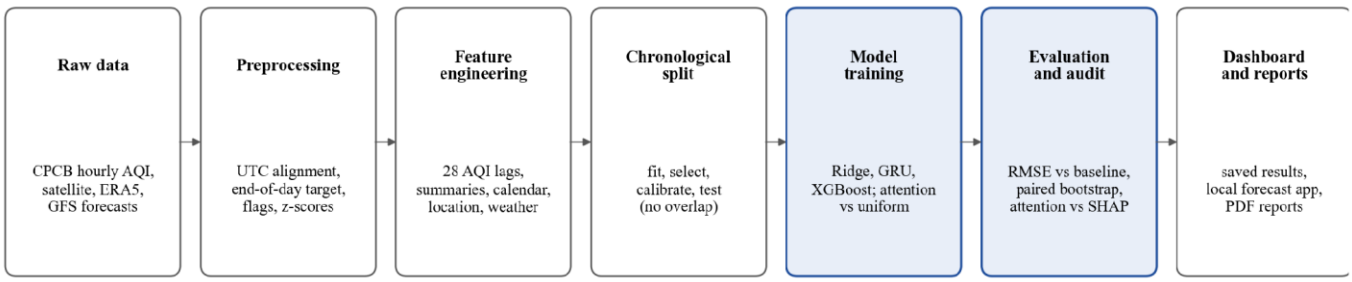


Fig. 1. System workflow from raw data to dashboard. The shaded stages are the study's two main experiments.

F. Experimental setup

TABLE VI
HARDWARE, SOFTWARE AND PARAMETERS

Item	Value
Computer	AMD Ryzen 7 7840HS (8 cores, 16 threads), 15.3 GB RAM, Windows 11 Home
GPU	NVIDIA GeForce RTX 4060 Laptop, 8 GB, driver 592.82, CUDA 12.8
Software	Python 3.13.3; PyTorch 2.10.0+cu128 [27]; XGBoost 3.4.1; scikit-learn 1.6.0 [28]; pandas 2.2.3; NumPy 2.1.2; SciPy 1.14.1; SHAP 0.53.0
GRU	1 layer, 64 units; Adam, learning rate 2×10^{-3}, weight decay 10^{-4}; batch 256; at most 60 epochs with early stopping (patience 8) on a 15% date-based validation split; gradient clip 1.0; MSE on the standardized target; 3 seeds (stopped at 23, 19 and 26 epochs)
XGBoost	600 trees, depth 5, learning rate 0.035, row sample 0.85, column sample 0.9, L2 5, minimum child weight 15, histogram method on GPU, seed 2
Ridge	alpha 1.0 after median imputation, missing indicators and standardization
Run time	Development grid for 2025 (all candidates, GPU): 4 min 21 s. CPU inference: GRU ensemble 65 ms and XGBoost 9 ms per 1,000 station-days

G. Evaluation protocol

For each forecast experiment, the model, the selection, and the radius of the interval were defined prior to the test period scoring (Table VII). There are different periods and different stations in three forecast experiments, and hence, each model is scored against the baseline independently on their respective rows. Attention experiments were conducted using three designs on the 2023 dataset: chronological (training period January-August, validating period September-October, testing period November-December), leave one city out in eight folds, and season blocks in four folds of alternated months. All the attention runs terminated early due to random holdout weeks; all the results provided here are from purged runs.

TABLE VII
DESIGN OF THE THREE LOCKED FORECAST TESTS

Step	2024 same-day	2025 next-day	2026 weather
Fit	Jan–Sep 2023	Jan 2023–Sep 2024	Jan 2024–Sep 2025
Select	Oct 2023	Oct 2024	Oct 2025
Calibrate	Nov–Dec 2023	Nov–Dec 2024	Nov–Dec 2025
Scored Rows	2024 28,803	2025 30,263	Jan–Sep 2026 20,086

The performance measures being applied are Root Mean Square Error (RMSE), Mean Absolute Error (MAE) and R^2. The improvements in performance measure are the difference between the RMSE of the two models in the same row, while the 95% confidence interval is calculated based on the paired calendar week-block bootstrap method (2,000 replications for year 2026 testing). The confidence interval assumes that the test week sampling is not adjusted for multiple comparison.A 90% prediction interval was specified as a fixed radius from the calibration errors, and its coverage was evaluated using the test data set. Only if the lower bound of the paired difference was positive for both the baseline and the GRU, and with no reduction in the sharp change detection performance, could a candidate model be considered as an alternative to the baseline.

H. Faithfulness audit

Two audits were run on frozen models. In the fixed-model audits, 300 input rows were sampled for each corrected FA and S-SA checkpoint. Inputs were replaced group by group with either the training mean or a same-city, same-season donor value, and the change in model output defined a sensitivity ranking. The Spearman correlation between that ranking and each method's ranking, attention weights and SHAP values (128 samples), was averaged over seeds and rows. In an exploratory audit of temporal attention, 1,000 rows were used to compare replacing the days attention ranked highest against random and reverse rankings. High agreement shows that a ranking follows the model's behaviour; it does not show physical causation.

I. Pseudocode

Algorithm 1. Locked forecast evaluation
Input: aligned table D; periods T_fit, T_sel, T_cal, T_test
Output: frozen model M_final, scores P, decision

```
D_c ← PREPROCESS(D)   // UTC day, AQI_eod, flags, lags
for each candidate m and feature family f do
    fit m on D_c[T_fit]; s_m ← RMSE(m, D_c[T_sel])
end for
M_final ← argmin s_m;  r ← 90th percentile of |y − M_final(x)| on D_c[T_cal]
FREEZE(M_final, r) and record file hashes
ŷ ← M_final(x) on D_c[T_test];  b ← previous-day AQI on D_c[T_test]
P ← {RMSE, MAE, R²} of ŷ and b on identical rows
for k = 1 to 2000 do
    resample calendar weeks; g_k ← RMSE(b) − RMSE(ŷ)
end for
CI ← 2.5th and 97.5th percentiles of g
promote M_final only if CI lower bound > 0 against b and the incumbent, with no sharp-change loss
return M_final, P, CI
```

Algorithm 2. Faithfulness audit of a frozen attention model
Input: model M, rows X, replacement rules R = {mean, seasonal donor}
Output: mean Spearman agreement of attention and SHAP

```
S ← SAMPLE(X, 300)
for each row x in S do
    a ← ATTENTION(M, x);  h ← SHAP(M, x)
    for each rule r in R do
        for each input group q do δ_q ← |M(REPLACE(x, q, r)) − M(x)|
        ρ_att ← SPEARMAN(a, δ);  ρ_shap ← SPEARMAN(|h|, δ)
    end for
end for
return mean ρ_att and ρ_shap over rows, averaged over seeds
```

IV. RESULTS AND DISCUSSION

A. Performance metrics

Each AQI-history model beat the baseline on its own rows (Table VIII) [29]. The gain was 3.39 AQI points (9.4%) for ridge with history in 2024, 2.87 (8.1%) for the GRU in 2025 and 4.40 (12.7%) for weather XGBoost in 2026, each with a 95% interval above zero. This indicates that the models forecast the next day more accurately than repeating today's value, but the typical error is still near 20 AQI points. Ridge-env, without

AQI history, had RMSE 63.239, 75% above the baseline, so inputs other than AQI history are not enough on their own. The nominal 90% ranges covered 94.7% to 95.4% of observations, so they are conservative.

TABLE VIII
LOCKED TEST RESULTS (RMSE, MAE IN AQI POINTS)

Model	Test	Rows	MAE	RMSE	R^2	Gain over baseline [95% CI]	Coverage
Baseline	2024 same-day	28,803	23.72	36.197	0.869	n/a	n/a
Ridge-env	2024 same-day	28,803	45.80	63.239	0.599	none	n/a
Ridge-history	2024 same-day	28,803	23.00	32.806	0.892	3.39 [2.03, 4.67]	94.7%
Baseline	2025 next-day	30,263	22.98	35.388	0.867	n/a	n/a
GRU-28days	2025 next-day	30,263	21.86	32.516	0.887	2.87 [2.17, 3.60]	95.4%
Baseline	2026 next-day	20,086	21.76	34.668	0.800	n/a	n/a
Frozen GRU-28days	2026 next-day	20,086	20.39	31.707	0.833	2.96	n/a
Weather XGBoost	2026 next-day	20,086	19.35	30.270	0.848	4.40 [3.19, 5.71]	94.9%

B. Comparison with baselines and ablations

A natural concern is that the GRU mostly repeats yesterday's AQI. To test this, the frozen GRU was re-run on all 2025 test rows with parts of its input removed (Table IX). Its predictions correlate 0.989 with yesterday's AQI and 68% lie within 10 points of it, so the concern is largely right: the GRU is anchored to yesterday. Yet yesterday's value alone is not enough. Given only yesterday's AQI, the GRU kept 12% of its gain; with yesterday repeated and calendar and location kept, 43%; and with the full history but no calendar or location, 55%. The ridge regression using all lags, date and place retains 78% of the gain, implying that the GRU adds merely 0.6 RMSE to the linear model. This is because the gain is derived from old days, seasons, and location, as well as by bringing extreme values closer to the mean (one variable model: y = 10.5 + 0.921 * yesterday). The features used in the removal experiment are unusual to the neural network, and hence overestimate its dependency on the omitted features.

TABLE IX
INPUT ABLATIONS OF THE 2025 GRU AND LINEAR COMPARISONS

Model	RMSE	Gain	Share of GRU gain
Baseline (yesterday)	35.388	0.00	0%
Linear, yesterday only	34.856	0.53	19%
Linear, three recent inputs	33.471	1.92	67%
Linear, all history, calendar, location	33.155	2.23	78%
GRU, only yesterday's AQI	35.044	0.34	12%
GRU, yesterday repeated	34.140	1.25	43%
GRU, no calendar or location	33.794	1.59	55%
GRU, older days shuffled	33.303	2.09	73%
GRU, full model	32.516	2.87	100%

Adding inputs step by step shows the same pattern for the tabular models fitted on 2023 data (Table X): only the AQI-history step beat the baseline, and station location helped within known cities but not on an unseen one. For held-out cities, the spatial S-TA+SA model reached a city-averaged RMSE of 54.16 against 57.19 for its uniform control and improved five of eight cities, whereas sequence TA did not improve on its control (55.23 against 55.11). This is cross-city accuracy and is not evidence about faithfulness.

TABLE X
BEST TEST RMSE BY INPUT SET, 2023 TABULAR MODELS

Inputs	Chrono.	Held-out city	Season
Satellite only	119.7	72.4	80.3
Weather only	97.8	60.8	60.7
Satellite + weather	93.5	58.5	57.2
+ station location	86.8	57.1	49.7
+ day of year, past weather	72.1	57.2	49.0
+ AQI history	42.4	31.3	34.9
Baseline (yesterday)	46.7	32.8	40.1

C. Feature importance of the weather model

The weather XGBoost uses 50 inputs (Table IV) and predicts the change from yesterday. On the 2026 test rows, the forecast weather carried 28% of the model's mean absolute SHAP value, close to yesterday's AQI at 29% and ahead of the older daily values at 24% (Fig. 2). Shuffling the weather inputs raised RMSE by 7.2 points, and by 5.9 when shuffled within each city, which keeps city climate. Setting the forecast to missing raised RMSE from 30.27 to 32.23, removing about 44% of the model's gain over the baseline. Forecast wind speed and humidity mattered most (RMSE rises of 3.2 and 2.5), which agrees with the physics of dispersion and moisture, but these scores describe the model's behaviour and do not prove cause. The development rows gave the same ranking. On those rows the history-only XGBoost scored close to the baseline, while adding weather lowered RMSE by 3 to 5 points (Table XI).

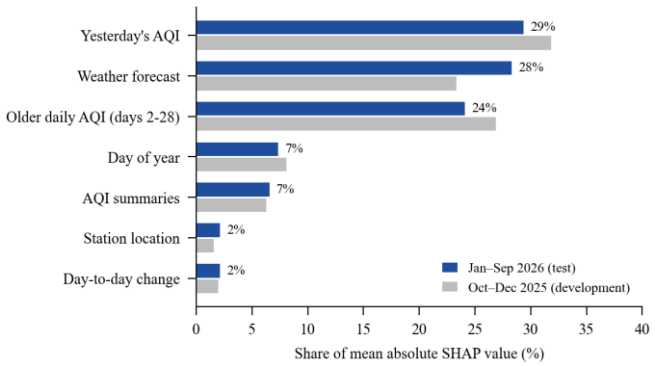


Fig. 2. Share of the weather XGBoost's mean absolute SHAP value by input group.

TABLE XI
WEATHER FORECAST ABLATION ON DEVELOPMENT ROWS (RMSE)

Model	Oct 2025	Nov–Dec 2025
Baseline	37.26	41.70
Frozen GRU	36.23	38.16
XGBoost, history only	36.78	40.25
XGBoost, with weather forecast	31.80	37.24

D. Attention and accuracy

Attention did not improve accuracy reliably (Fig. 3). Across the 36 paired comparisons, attention helped in 11, hurt in 11 and was inconclusive in 14 [30]. The largest reversal was S-SA in the chronological test, which gained 9.49 on SWT-w7 and lost 3.58 on SW-w7. The most consistent result was S-TA+SA in held-out cities, which gained 3.03 on SWT-w7 and 3.00 on SW-w7 but showed no chronological gain on either set. Feature attention alone was a clear gain once and a clear loss in four of its six comparisons. With 36 unadjusted intervals, a few clear results would be expected by chance. These models omit AQI history, so their RMSE (about 53 to 95) is far above the forecast models'.

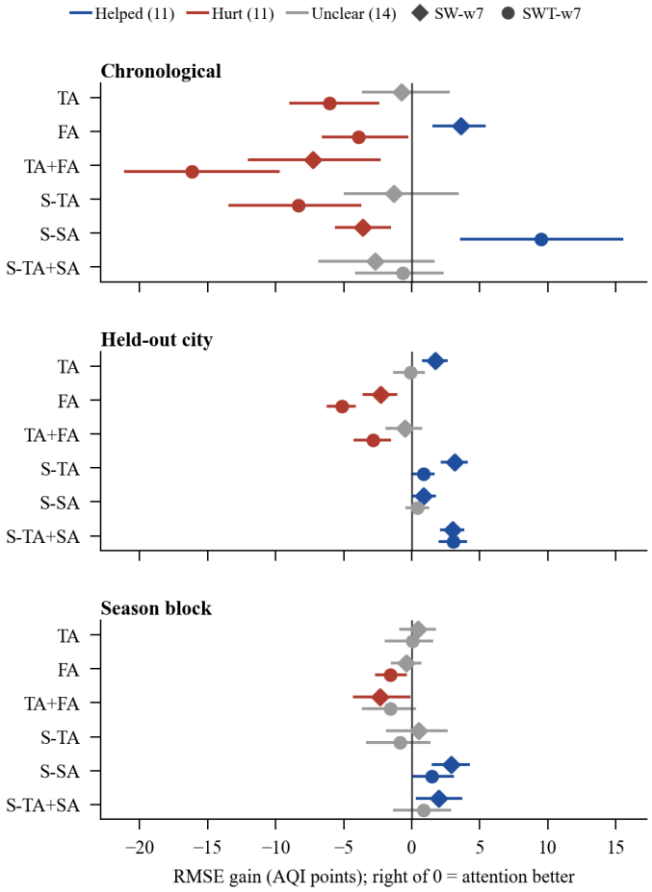


Fig. 3. RMSE gain of each attention variant over its uniform control, with 95% intervals, on two input sets.

E. Faithfulness of attention

In all four fixed-model audits attention agreed less than SHAP with replacement sensitivity (Table XII), which suggests that attention is a weaker guide to what the model uses. The result varied by city and replacement method. Under mean replacement, spatial attention nearly matched SHAP in Bengaluru (0.673 against 0.677) but was near zero or negative in Delhi (0.100), Jaipur (−0.017) and Lucknow (−0.104). City samples are uneven, from one row in Mumbai to 148 in Delhi, so small-city figures are weak.

TABLE XII
RANK AGREEMENT WITH REPLACEMENT SENSITIVITY (SPEARMAN)

Attention type	Replacement	Attention	SHAP
Feature (FA)	Training mean	0.420	0.664
Feature (FA)	Seasonal donor	0.392	0.546
Spatial (S-SA)	Training mean	0.242	0.419
Spatial (S-SA)	Seasonal donor	0.483	0.586

The exploratory temporal-attention audit reveals that the weights are not hollow. The difference in performance when removing the highest ranked day attention compared to a random day and the lowest ranked day attention was 37.1 versus 17.3 and 7.5 AQI points respectively. The rank correlation with leave-one-day-out sensitivity analysis was 0.488, and this number did not drop even on the polluted days, being 0.547 for AQI 300+ (320 rows), and 0.461 for below (680 rows). The worst city was Chennai, with 0.178.

F. Error analysis

All models read too high on clean days and too low on polluted days (Fig. 4). The Severe band is only 2.4% of station-days, so the models see few examples of it and regress toward the mean. In the 401–500 band (94 station-days in 2026) the mean error was −44 for the weather model and −72 for the GRU, which means none of the models is suited to alerting on the worst days. Sharp changes are also missed (Table XIII). The weather model detected 18.2% of moves of 50 points or more in the correct direction against 6.7% for the GRU, but its precision fell from 57.5% to 52.8%, so it failed the promotion rule's no-loss condition despite a lower RMSE (paired gain over the GRU 1.44, 95% interval 0.38 to 2.61). On days that changed by less than 25 points, the baseline had the lowest error, so the models earn their gain on the days that change.

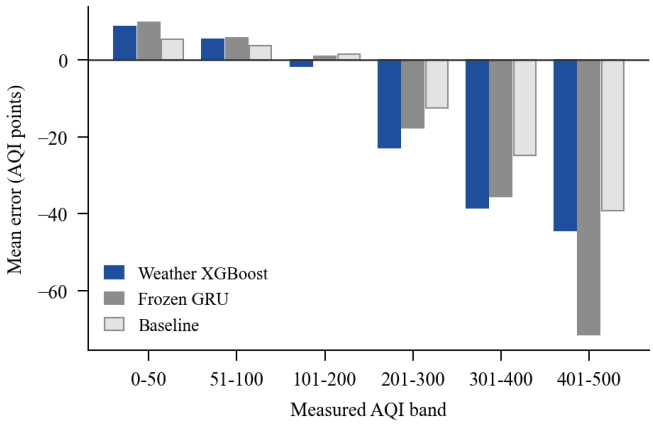


Fig. 4. Mean signed error by measured AQI band, January–September 2026. Below zero means too low.

TABLE XIII
RMSE BY SIZE OF ACTUAL DAILY AQI CHANGE, 2026

Change (points)	Days	Weather	GRU	Baseline
Under 25	14,202	15.98	13.82	11.05
25 to under 50	3,569	33.00	33.52	35.57
50 or more	2,315	68.59	76.28	87.91

By city, the GRU improved on the baseline in all eight cities in 2025 (Fig. 5), with the largest gain in Delhi (3.74) and the smallest in Mumbai (1.29), although Delhi's error stays the largest at 41.5. Two alert classifiers fit on October 2024 and scored once on 2025 did not pass their promotion checks and were not deployed: for AQI of 300 or more the GRU threshold had precision 83.7% and recall 82.9% against 88.4% and 64.9% for the classifier.

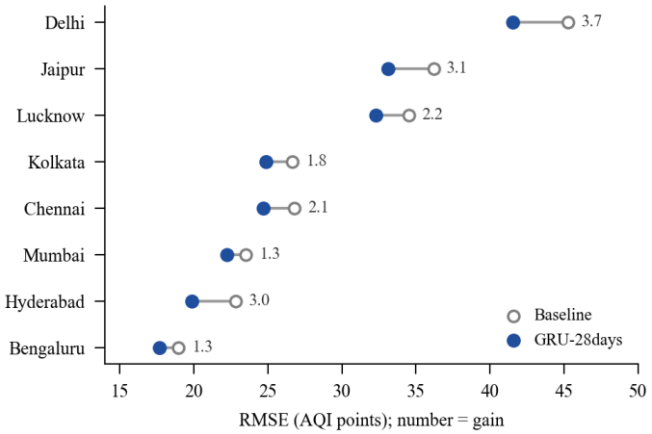


Fig. 5. RMSE of the GRU and the baseline by city, 2025 test (30,263 station-days). Numbers show the gain.

G. Interpretation of results

Three conclusions follow. First, forecast skill beyond the baseline is real but modest: 8% to 13% in RMSE. Most of what the models predict is yesterday's value, the history-only GRU adds little over a linear model, and the clearest additional signal comes from weather forecasts. Second, attention is neither a reliable accuracy gain nor, in the sampled audits, as faithful a guide to model behaviour as SHAP, and its effects depend on the input set, the test design and the city. An attention map should therefore be checked for each model and setting before it is read as an explanation. Third, the forecasts regress toward the middle, which limits their use on the days that matter most for health. These findings describe model behaviour on past data and do not identify physical causes of pollution.

H. Comparison with past literature

The reviewed studies report scores on different targets, horizons and scales, so this work does not rank its numbers against theirs (Table XIV). The comparison below is therefore about what each study shows, and it connects the results to the gaps in Section II.

TABLE XIV
REPORTED RESULTS OF REVIEWED STUDIES AND THEIR RELATION TO THIS WORK

Study	Reported result	Relation to this work
Chen et al. [6]	Hourly PM2.5, RMSE 9.82 one step and 13.49 averaged over 1–48 h	Different pollutant, horizon and scale. Attention is a predictive precedent there; here its accuracy effect varied (11 helped, 11 hurt, 14 unclear).
Liu et al. [7]	PM2.5 MAE 13.97, 16.13 and 4.07 on three datasets	Component ablation there tests architecture; here a fixed-model audit tests whether attention reflects model behaviour.
Ma et al. [5]	Lower error with four frozen layers than with one after target-site fine-tuning (scaled units)	Adaptation uses target labels. The locked tests here use none.
Li et al. [19]	Site-based PM10 R^2 about 0.85 on 288 held-out stations	Spatial assessment, not forecasting. Next-day R^2 here was 0.887 (2025) on a different target.
Liu et al. [14]	Beijing daily AQI RMSE 3.0012	Different scale and preprocessing, and split independence is unverified there.
Zhang et al. [10]	Intrinsic explanations agreed closely with GradientSHAP	Here attention agreed less than SHAP with replacement sensitivity in four of four audits. Models and cities differ, so the contrast is indicative.
Turbé et al. [8]	Shapley sampling ranked best among six attribution methods	Consistent in direction: SHAP agreed more with measured sensitivity than attention did.
Šić et al. [9]	Suitable perturbation depends on dataset and model	Consistent: S-SA agreement moved from 0.242 to 0.483 when the replacement rule changed.

V. CONCLUSION

In this paper, we attempted to quantify the improvement of AQI and weather models relative to a previous day baseline and whether attention helps us understand why models work. We created an aligned dataset of 107 stations in 8 cities and conducted three locked forecast experiments. The GRU improved RMSE from 35.388 to 32.516 in 2025, and the weather XGBoost improved it from 34.668 to 30.270 in 2026, while the history-only GRU was marginally better than a linear model and the weather forecast accounted for most of the improvement. In 36 trials, attention failed to significantly increase accuracy and in every sampled audit it disagreed with SHAP more than the actual sensitivity measure. The contribution of this work is a reproducible analysis that distinguishes forecast quality, input importance, and explanation fidelity with a dashboard of the saved results. Our conclusions are that attention visualizations need to be tested

before being used to explain environmental forecasts and that the previous day baseline should go hand-in-hand with accuracy reporting.

VI. Study Limitations

Dataset limitations. The dataset includes exports from CPCB station exports as well as public satellite, reanalysis and forecast data for eight Indian cities. There is inconsistency in station availability, because Delhi provides 39 out of 107 stations, and 16 stations were left unscored in 2026 due to non-availability or lack of target in export data. Satellites have limitation due to cloud cover, while weather forecasts are available on the city level and with unknown delivery lag. Attention analysis relies solely on 2023 data.

Technical and environmental limitations. All experiments were conducted on one laptop with 8 GB GPU. Three seeds were used for attention experiment, while in the case of fixed-model audits the number of sampled rows was 300 (1,000 for the temporal-attention audit), thus the intervals represent sampling within a test week and not seed variability. Input replacement with a training mean can result in unrealistic inputs, while correlated inputs get credit in importance scores.

Scope limitations. The evaluation was offline on historical data and did not include live forecasts, because no same-day AQI source is configured. The out-of-time evaluation for 13 October to 31 December 2026 has no score as of 2 November 2026. The audits are model-behaviour interventions and not causal tests of atmospheric processes. LIME, remove-and-retrain, concept-activation testing and an audit of the held-out-city checkpoints were outside the scope of this study.

VII. Future Scope

Recommended Enhancements. The frozen mixture of 30% GRU and 70% weather XGBoost model ought to be assessed based on the out-of-time assessment post 1 January 2027 using the promotional criterion provided in Section III. The historical only XGBoost model can be re-trained and evaluated on the 2026 test to determine the impact of weather alone. The audit process may be expanded to include held-out city checkpoints and LIME, Remove and retrain, and concept-based stagnation test. Tail-oriented training criteria will be able to tackle the under forecasting of the Severe band category.

Real-time implementation. A same-day AQI source and a live weather forecast feed would allow forecasts to be recorded before their target day, so that live skill can be measured and not only backtested. The out-of-time status is tracked in [31].

Extensions of the approach. The methodology can be extended to work with PM2.5 and PM10 objectives, other cities, and health advisories based on the dashboard. The audit process may also be used to assess attention-based explanations in other environmental forecasting contexts.

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References

[1] World Health Organization, “Ambient (outdoor) air pollution,” Fact sheet. [Online]. Available: https://www.who.int/news-room/fact-sheets/detail/ambient-(outdoor)-air-quality-and-health

[2] State of Global Air, “State of Global Air Report 2024,” Health Effects Institute, 2024. [Online]. Available: https://www.stateofglobalair.org/resources/report/state-global-air-report-2024

[3] India State-Level Disease Burden Initiative Air Pollution Collaborators, “Health and economic impact of air pollution in the states of India: The Global Burden of Disease Study 2019,” *Lancet Planet. Health*, vol. 5, no. 1, pp. e25–e38, 2021, doi: 10.1016/S2542-5196(20)30298-9.

[4] Central Pollution Control Board, “National Air Quality Index and AQI data repository.” [Online]. Available: https://airquality.cpcb.gov.in/AQI_India/

[5] Z. Ma *et al.*, “Air pollutant prediction model based on transfer learning two-stage attention mechanism,” *Sci. Rep.*, vol. 14, no. 1, Art. no. 7385, 2024, doi: 10.1038/s41598-024-57784-7.

[6] C. Chen *et al.*, “Prediction of pollutant concentration based on spatial–temporal attention, ResNet and ConvLSTM,” *Sensors*, vol. 23, no. 21, Art. no. 8863, 2023, doi: 10.3390/s23218863.

[7] H. Liu, Q. Han, H. Sun, J. Sheng, and Z. Yang, “Spatiotemporal adaptive attention graph convolution network for city-level air quality prediction,” *Sci. Rep.*, vol. 13, no. 1, Art. no. 13335, 2023, doi: 10.1038/s41598-023-39286-0.

[8] H. Turbé, M. Bjelogrlic, C. Lovis, and G. Mengaldo, “Evaluation of post-hoc interpretability methods in time-series classification,” *Nature Mach. Intell.*, vol. 5, no. 3, pp. 250–260, 2023, doi: 10.1038/s42256-023-00620-w.

[9] I. Šić, E. Veas, and V. Sabol, “A comprehensive analysis of perturbation methods in explainable AI feature attribution validation for neural time series classifiers,” *Sci. Rep.*, vol. 15, no. 1, Art. no. 26607, 2025, doi: 10.1038/s41598-025-09538-2.

[10] Z. Zhang, D. Schlesinger, C. Johansson, M. Engardt, and X. Ma, “X2-AQFormer: Unveiling dynamic drivers in multi-day hourly air pollution forecasting,” *npj Clean Air*, vol. 2, no. 1, Art. no. 17, 2026, doi: 10.1038/s44407-026-00058-5.

[11] AQI project, “Literature survey,” docs/AQI_Literature_Survey_corrected.docx, Sep. 2026.

[12] S. Jain and B. C. Wallace, “Attention is not explanation,” in *Proc. NAACL-HLT*, 2019, doi: 10.18653/v1/N19-1357.

[13] S. Wiegreffe and Y. Pinter, “Attention is not not explanation,” in *Proc. EMNLP-IJCNLP*, 2019, doi: 10.18653/v1/D19-1002.

[14] X. Liu, K. Su, S. Wang, and K. H. Ghazali, “Intelligent prediction of air quality index based on the Transformer-BiLSTM model,” *Sci. Rep.*, vol. 15, no. 1, Art. no. 41838, 2025, doi: 10.1038/s41598-025-25865-w.

[15] D. Bahdanau, K. Cho, and Y. Bengio, “Neural machine translation by jointly learning to align and translate,” in *Proc. ICLR*, 2015, arXiv:1409.0473.

[16] S. M. Lundberg and S.-I. Lee, “A unified approach to interpreting model predictions,” in *Proc. NeurIPS*, 2017, arXiv:1705.07874.

[17] S. Krishna *et al.*, “The disagreement problem in explainable machine learning: A practitioner’s perspective,” arXiv:2202.01602, 2022.

[18] M. Hertel, S. Pütz, R. Mikut, V. Hagenmeyer, and B. Schäfer, “Explainable time-series forecasting with sampling-free SHAP for Transformers,” *Nature Commun.*, vol. 17, no. 1, Art. no. 4723, 2026, doi: 10.1038/s41467-026-73243-5.

[19] L. Li *et al.*, “Improving air quality assessment using physics-inspired deep graph learning,” *npj Clim. Atmos. Sci.*, vol. 6, no. 1, Art. no. 152, 2023, doi: 10.1038/s41612-023-00475-3.

[20] M. Poelzl, R. Kern, S. Kecorius, and M. Lovrić, “Exploration of transfer learning techniques for the prediction of PM10,” *Sci. Rep.*, vol. 15, no. 1, Art. no. 2919, 2025, doi: 10.1038/s41598-025-86550-6.

[21] J. P. Veefkind *et al.*, “TROPOMI on the ESA Sentinel-5 Precursor: A GMES mission for global observations of the atmospheric composition for climate, air quality and ozone layer applications,” *Remote Sens. Environ.*, vol. 120, pp. 70–83, 2012.

[22] NASA LP DAAC, "MCD19A2.061 Terra and Aqua MAIAC land aerosol optical depth daily 1 km," Google Earth Engine Data Catalog. [Online]. Available: https://developers.google.com/earth-engine/datasets/catalog/MODIS_061_MCD19A2_GRANULES

[23] J. Muñoz-Sabater *et al.*, "ERA5-Land: A state-of-the-art global reanalysis dataset for land applications," *Earth Syst. Sci. Data*, vol. 13, no. 9, pp. 4349–4383, 2021.

[24] Open-Meteo, "Previous Runs API." [Online]. Available: https://open-meteo.com/en/docs/previous-runs-api

[25] T. Chen and C. Guestrin, "XGBoost: A scalable tree boosting system," in *Proc. 22nd ACM SIGKDD Int. Conf. Knowl. Discovery Data Mining*, 2016, doi: 10.1145/2939672.2939785.

[26] K. Cho *et al.*, "Learning phrase representations using RNN encoder-decoder for statistical machine translation," in *Proc. EMNLP*, 2014, doi: 10.3115/v1/D14-1179.

[27] A. Paszke *et al.*, "PyTorch: An imperative style, high-performance deep learning library," in *Proc. NeurIPS*, 2019.

[28] F. Pedregosa *et al.*, "Scikit-learn: Machine learning in Python," *J. Mach. Learn. Res.*, vol. 12, pp. 2825–2830, 2011.

[29] AQI project, "Locked evaluation records," results/year_holdout/runs/2023-to-2024-v1, results/forecast_2025/runs/locked-2025-v1 and results/forecast_2026/runs/locked-2026-v1 (metrics, manifests and verification records), 2026.

[30] AQI project, "Attention versus uniform control across input sets and protocols," results/attention_stability/gains.csv, generated by src/models/report_attention_stability.py, Nov. 2026.

[31] AQI project, "Prospective tracking status," results/prospective/status.json, updated 2 Nov. 2026.